EXPERIMENT - 4 (8-BIT DIVISION)

Aim: To write and assembly language program to implement 8-bit division using 8085 processor.

Algorithm:

1. Start the program by loading a register pair with the address of memory location.
2. Move the data to a register.
3. Get the second data and load it into the accumulator.
4. Subtract the two register contents.
5. Increment the value of the carry.
6. Check whether the repeated subtraction is over.
7. Store the value of quotient and remainder in the memory location.
8. Halt.

PROGRAM:

MNEMONICS	EXPLANATION
START: NOP	It is often used for code alignment.
LDA 8500	Load the accumulator with first number in address 8500
MOV B,A	Move data from accumulator to 'B' register.
LDA 8501	Load the accumulator with 2nd number in the address 8501
MVI C,00H	Move the immediate value 00 into register 'C'.
LOOP: CMP B	LOOP: label for loop; CMP B: Compare the value in accumulator (A) with (B)
JC LOOP1	If the carrying is (A<B) jump to label loop1.
SUB B	Subtract the value in register (B) from the accumulator.
INR C	Increment register C by 1.
JMP LOOP	Jump back to the begining of the loop.

STA 8502	Store the data in accumulator
MOV A, C	Move the data from 'c' register to accumulator.
STA 8503	Store the data in the accumulator for in 8503.
RST 1	Typically transfer control.
HLT	HALT.

INPUT:

ADDRESS	DATA
8500	2
8501	6

OUTPUT:

ADDRESS	DATA.
8502	0
8503	3

RESULT:

Thus the program was executed successfully using 8085 processor simulator.


GNUSim8085 - 8085 Microprocessor Simulator
File Reset Assembler Debug Help
Registers
A 03
BC 00 00
DE 00 05
HL 00 0F
PSW 00 00
PC 42 16
SP FF FF
Int-Reg 00
Flag
S 0
Z 1
AC 0
P 1
C 0
Load me at
LDA 2200
MOV E,A
MVI D,00
LDA 2201
MOV C,A
LXI H,0000
BACK: DAD D
DCR C
JNZ BACK
SHLD 2202
HLT
Decimal - Hex Convertion
Decimal
Hex
0
0
To Hex
To Dec
I/O Ports
0
00
Update Port Value
Memory
2201
03
Update Memory
Data
Stack
KeyPad
Memory
I/O Ports
Start 2202
OK
Address (Hex) Address Data
089A 2202 15
089B 2203 0
089C 2204 0
089D 2205 0
089E 2206 0
089F 2207 0
08A0 2208 0
08A1 2209 0
08A2 2210 0
08A3 2211 0
08A4 2212 0
08A5 2213 0
08A6 2214 0
08A7 2215 0
08A8 2216 0
08A9 2217 0
Line No Assembler Message
0 Program assembled successfully